



Metropolitan University, Sylhet

Department of Computer Science & Engineering

Lab Report – 06:

**Longest Common Subsequence (LCS) using
Dynamic Programming in C**

Course Title: Algorithm Design and Analysis Lab

Course Code: CSE 232

Submitted To:

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problem:

Longest Common Subsequence (LCS) using Dynamic Programming in C

CODE :

```
// Longest Common Subsequence (LCS) using Dynamic Programming in C
// Name: Mahdi Hasan Shuvo | ID: 251-115-030 | Batch: CSE 62nd (A)

#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

void LCS(char str1[], char str2[]) {
    int len1 = strlen(str1);
    int len2 = strlen(str2);
    int dp[len1 + 1][len2 + 1];
    for (int i = 0; i <= len1; i++) {
        for (int j = 0; j <= len2; j++) {
            dp[i][j] = 0;
        }
    }
    for (int i = 1; i <= len1; i++) {
        for (int j = 1; j <= len2; j++) {
            if (str1[i - 1] == str2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }
    int lcsLen = dp[len1][len2];
    char lcs[lcsLen + 1];
    lcs[lcsLen] = '\0';
    int i = len1, j = len2;
    while (i > 0 && j > 0) {
        if (str1[i - 1] == str2[j - 1]) {
            lcs[lcsLen - 1] = str1[i - 1];
            lcsLen--;
            i--;
            j--;
        }
    }
    lcs[lcsLen] = '\0';
    printf("Longest Common Subsequence: %s\n", lcs);
}
```

```

        j--;
    } else if (dp[i - 1][j] > dp[i][j - 1]) {
        i--;
    } else {
        j--;
    }
}

printf("LCS: %s\n", lcs);
}

int main() {
    char str1[100], str2[100];

    printf("Enter string-1: ");
    scanf("%s", str1);

    printf("Enter string-2: ");
    scanf("%s", str2);

    LCS(str1, str2);

    return 0;
}

```

Output Output Output

OUTPUT:

Output
Enter string-1: ABCBDAB Enter string-2: BDCABA LCS: BDAB === Code Execution Successful ===

Thank You sir